

MEWS Development Program

Solution & Design — Design and Development Reference

Source basis: Design and Development Plan (MSTI/TECH/002/111024/00) and Design and Verification Report (MSTI/TECH/004/111024/00).

Evidence status: Portfolio reference based on the supplied project forms. It is not a controlled reproduction of the original records.

1. Design & Development Framework

The supplied Design and Development Plan defines a structured sequence for moving from planning and design inputs through design review, design outputs, manufacturing readiness, verification, external review, and validation.

1. Plan	2. Design Input	3. Regulatory Review
4. Internal Design Review	5. Design Output	6. Manufacturing Capability
7. Design Verification	8. External Design Review	9. Design Validation

2. Design Inputs & Requirements

The plan identifies design inputs in terms of **functional and performance requirements**, together with past experience and other requirements. Statutory and regulatory requirements are separately identified and reviewed as part of the development process.

3. Design Reviews & Outputs

The framework includes an **internal design review** followed by defined design outputs. The supplied form identifies outputs including **drawings, bill of material (BOM), and specifications**. These outputs provide the basis for subsequent manufacturing and verification activities.

4. Manufacturing Readiness

Before or alongside verification activities, the plan considers manufacturing capability, release of manufacturing and purchasing data, process adequacy, and equipment adequacy. This connects design decisions with practical production readiness.

5. Design Verification

The supplied Design and Verification Report records the relationship between **design input requirements** and the **design output observed**, followed by a conclusion and remarks. The form also provides preparation, review, and management acknowledgement fields.

Design Input Requirements	Design Output Observed	Conclusion / Remarks
Defined requirement	Observed product/design output	Verification conclusion

Prepared by	Noted by	Acknowledged by
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6. External Review & Validation

The Design and Development Plan provides for **external design review**, including customer review where required and clarification with purchasing and manufacturing personnel. It then defines **design validation** through product testing on the defined application by field trials under operating conditions.

7. Technical Project Management Perspective

From a Technical Project Management perspective, this framework provides a clear control path for coordinating requirements, design activities, engineering outputs, manufacturing dependencies, verification evidence, external stakeholders, and validation readiness.

The PM focus is on keeping these activities aligned, visible, scheduled, reviewed, and connected to project risks and delivery decisions — while the detailed engineering design and verification work remains with the appropriate technical personnel.

8. Evidence Boundary

This portfolio reference does not reproduce proprietary drawings, BOM part numbers, detailed specifications, controlled design records, confidential supplier information, patient information, or other restricted engineering material. The supplied forms are used as the structural basis for the portfolio presentation.

Portfolio Reference — anonymized solution and design overview. Not a controlled QMS or project record.